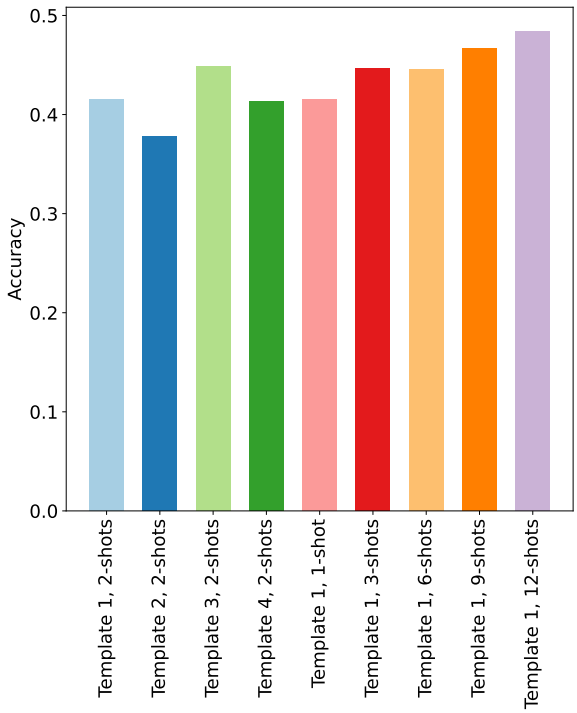

Figure 25: Accuracy of different prompts on the emotion classification task. This experiment varies both the prompt template and the number of provided few-shot examples. Experiments are conducted on Llama 3.2 1B.

E Appraisal Probing

As detailed in Section 7 and similar to emotion probing experiments provided in Section 5, we perform probing experiments to assess the presence and strength of appraisal signals at different activations within the Llama 3.2 1B model. We perform probing separately for each appraisal dimension over different activation locations and layers across the model for the last token. The probing results in Figure 29 are measured as the regression R^2 score on a held-out test set.

Following the behavior observed in probing emotion signals, models begin consolidating appraisal-related information in the hidden states $\mathbf{h}^{(l)}$ around the mid-layers. We observe that beyond layer 10, there is no significant increase in probe accuracy in any appraisal dimension. As also observed in the emotion probing experiment, there is no clear distinction in probing performance between $\mathbf{m}^{(l)}$ and $\mathbf{h}^{(l)}$.

As discussed in length in the main test, the success of linear probing highly depends on whether the target concept is linearly detectable given an activation. The results here also enforce the notion that the appraisal signals are not linearly detectable at earlier layers but are strongly present as we approach the hidden state of the final layers.

F Further Details on Appraisal Modulation

As discussed in Section 9, we show the possibility of indirectly modifying the emotion of an input example by modulating its appraisals within the model representations. In this section, we provide more details and further experiments on appraisal modulation.

First, we redefine our appraisal modulation method to generalize to cases where we want to modify multiple concepts simultaneously. Let's assume that we have a set of r appraisal vectors $\mathcal{A} := \{\mathbf{v}_{i_1}, \mathbf{v}_{i_2}, \cdots, \mathbf{v}_{i_r}\}$ which we want to modify, and consider a second set $\mathcal{B} := \{\mathbf{v}_{j_1}, \mathbf{v}_{j_2}, \cdots, \mathbf{v}_{j_k}\}$ to contain the k appraisal vectors which we want to maintain the corresponding appraisal concept fixed during the update. We define the project matrix $\mathbf{P}_{\mathcal{B}}$ as the projection matrix which projects into the span of $\mathcal{B}$. More formally, form the matrix $\mathbf{V}_{\mathcal{B}} = [\mathbf{v}_{j_1}; \mathbf{v}_{j_2}; \cdots; \mathbf{v}_{j_k}] \in \mathbb{R}^{d \times k}$ by concatenating the vectors in $\mathcal{B}$ as the columns of $\mathbf{V}_{\mathcal{B}}$. With this in mind, the projection matrix, $P_{\mathcal{B}}$ as

$$P_{\mathcal{B}} = \mathbf{V}_{\mathcal{B}}(\mathbf{V}_{\mathcal{B}}^{\top}\mathbf{V}_{\mathcal{B}})^{-1}\mathbf{V}_{\mathcal{B}}^{\top},$$

and define the net effect vector as

$$\mathbf{z}_{\mathcal{A}} := (I - \mathbf{P}_{\mathcal{B}}) \sum_{a=i_1}^{i_r} \gamma_a \mathbf{v}_a,$$

where each γ_a is a variable from $\{-1, +1\}$ to indicate if the modulation promotes concept a or demote it. Finally, the modulation is performed as

$$\mathbf{x} \leftarrow \mathbf{x} + \beta \frac{\mathbf{z}_{\mathcal{A}}}{\|\mathbf{z}_{\mathcal{A}}\|_2}$$

where $\beta \in \mathbb{R}_+$ is a positive scaling factor.After performing the intervention, we measure the intervention's success by evaluating the new emotion label obtained by this modification across all examples in the dataset.

Section 9 reported the inference-time intervention results targeting the hidden state at layer 9 in Llama 3.2 1B, as we showed it is a critical point in emotion processing in Llama 3.2 1B. Here, we report the intervention results across all layers for a varied set of appraisal dimensions and their superposition to create more complex but specific concepts.

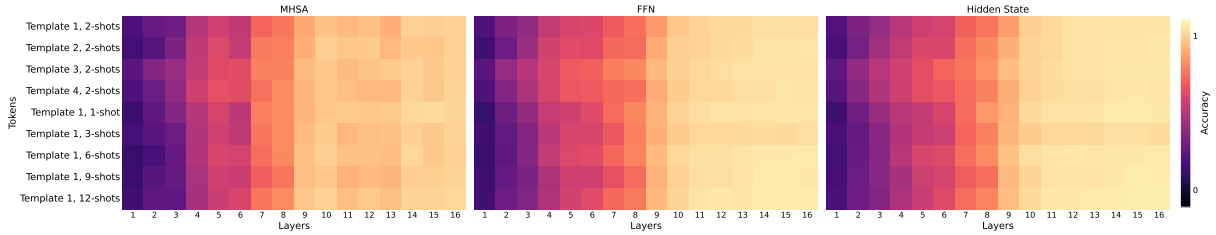

Figure 26: Probing accuracy with different prompts measured at last token in Llama 3.2 1B for all layers. This experiment varies both the prompt template and the number of provided few-shot examples.

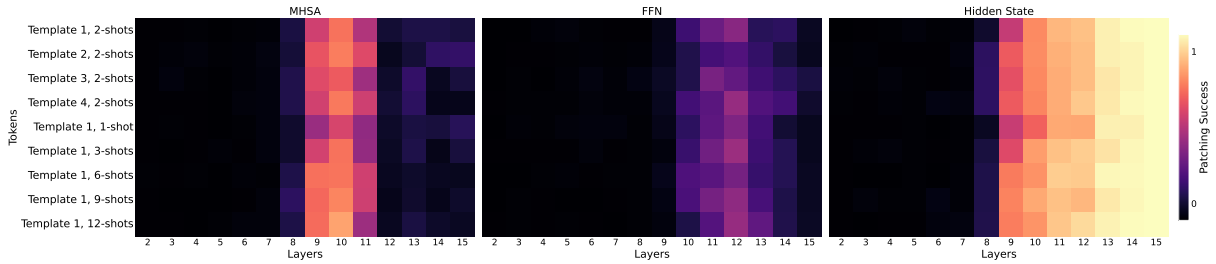

Figure 27: Success of activation patching with different prompts, measured at the last token index in Llama 3.2 1B across all layers with span = 3. This experiment varies both the prompt template and the number of provided few-shot examples.

As shown in Figures 30, 31, and 32, the intervention on appraisal concepts changes the distribution of the output labels and this distribution shift is intensified with higher values of β in all intervention experiments. However, the shift in the distribution of represented emotions does not necessarily conform with theoretical and intuitive expectations when intervening on earlier layers, particularly noticeable when β is sufficiently large. For example, in Figure 30, we note an unexpected decrease in the distribution of *joy* and *pride* in early layers, whereas psychologically plausible manipulations—such as an increase in high-valence emotions like *joy*, *pride*, and *surprise*, only emerge in mid-layers, peaking at layer 9. This observation supports the notion that intervening on the first layers is not effective because the linear structure in representations is not formed well yet.

Observing the intervention effect on later layers, we see significantly less pronounced distribution shifts. This also supports our earlier finding that the intervention on final layers is not effective because of the orthogonality of concepts that we showed in Section 8.

On the other hand, We observe a remarkable alignment with theoretical and intuitive expectations in the distribution shifts associated with interventions on middle layers (specifically layers 9-11). For instance, we observe that increasing the *pleasantness* appraisal promotes both *joy* and *pride*, aligning with the fact that both of these emotions have high associations with the appraisal. Also see Figure 35 for a different visualization.

To better evaluate intervention success, we also provide intervention results on the superposition of two appraisal dimensions (e.g., *other-agency* and *pleasantness*) across all layers in Llama 3.2 1B. The results, demonstrated in Figure 33, show a successful promotion of emotion *pride* with no further occurrences of *joy* in layers 9-11 when demoting *other-agency* and the promotion of *guilt* and *fear* with no occurrence of anger in mid-layers when demoting *other-agency*. Similarly, in Figure 34, we see the transition from *pride* to *surprise* and a larger distribution of *fear* as compared with *anger* when we promote *unpredictability*.

Finally, we will conclude this section by providing a control experiment in which, instead of using appraisal vectors, we intervene in the activation using a random vector. The results of this experiment are provided in Figure 36, and we see that no psychologically valid pattern is observable. Overall, these findings provide strong evidence that the mid-layers in Llama 3.2 1B meaningfully and directly contribute to cognitive processes related to emotions.

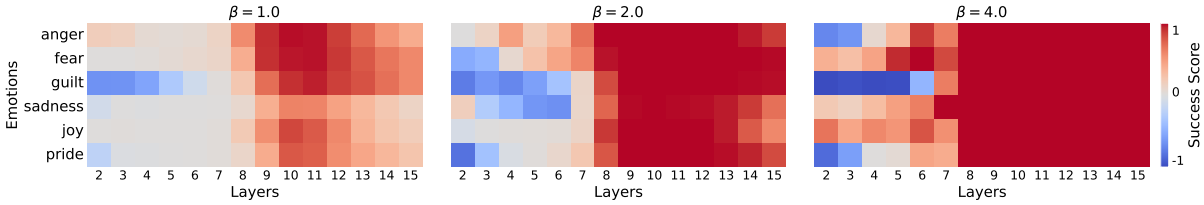

Figure 28: The heatmap showing the success of direct emotion promotion when applied at different layers of Llama 3.2 1B. Success score 1 means that the intervention successfully changed all output labels to the target emotion label. Scores 0 means that the intervention made no changes to the output and -1 means that the intervention resulted to complete opposite results, even damaging the samples with the correct original label. All the interventions in this figure used an intervention of layer span size 3.

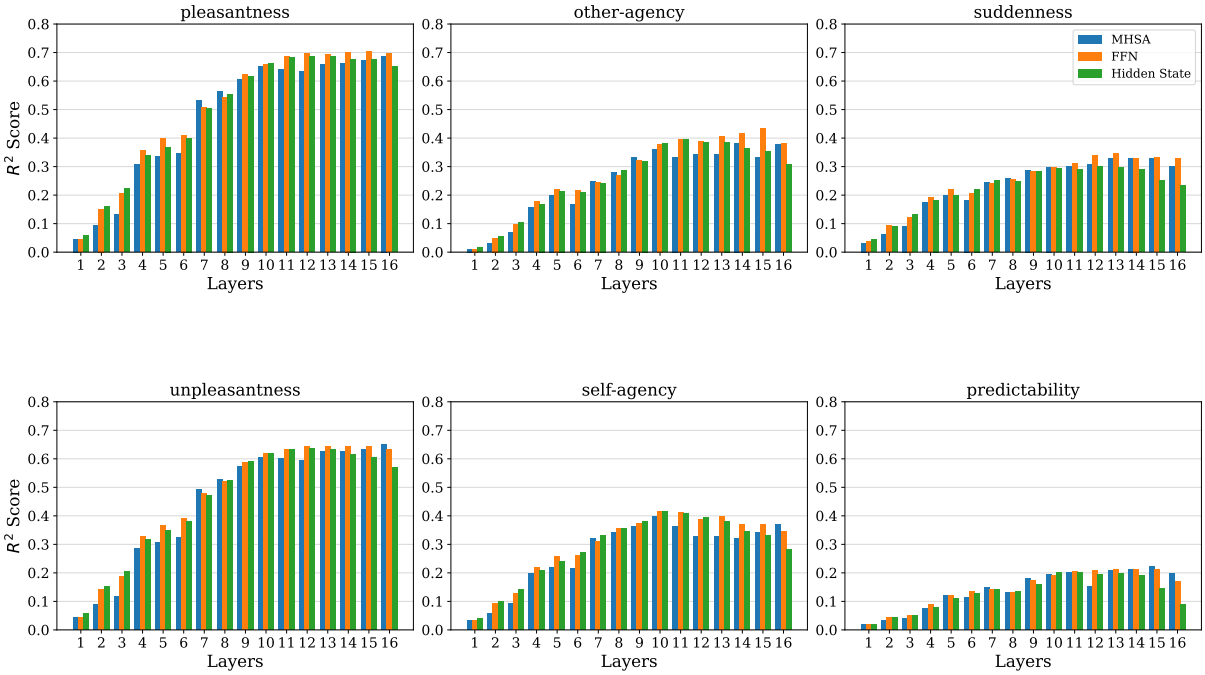

Figure 29: Probing results for Llama 3.2 1B, conducted separately for each appraisal dimension across different activation locations and layers for the last token. Results are measured as the regression R^2 score on a held-out test set.

G Code and Compute Resources

Our experiments are conducted using GPU-accelerated compute resources, with hardware such as NVIDIA A100 GPUs. For larger models, our studies are feasible on GPUs with at least 40GB of VRAM, with the full experiment running in approximately 24 hours. For smaller models, GPUs with 12GB of VRAM are sufficient to carry out our analyses efficiently.

Generative AI tools are utilized to improve the tone and style of writing, as well as for code completion during implementation.